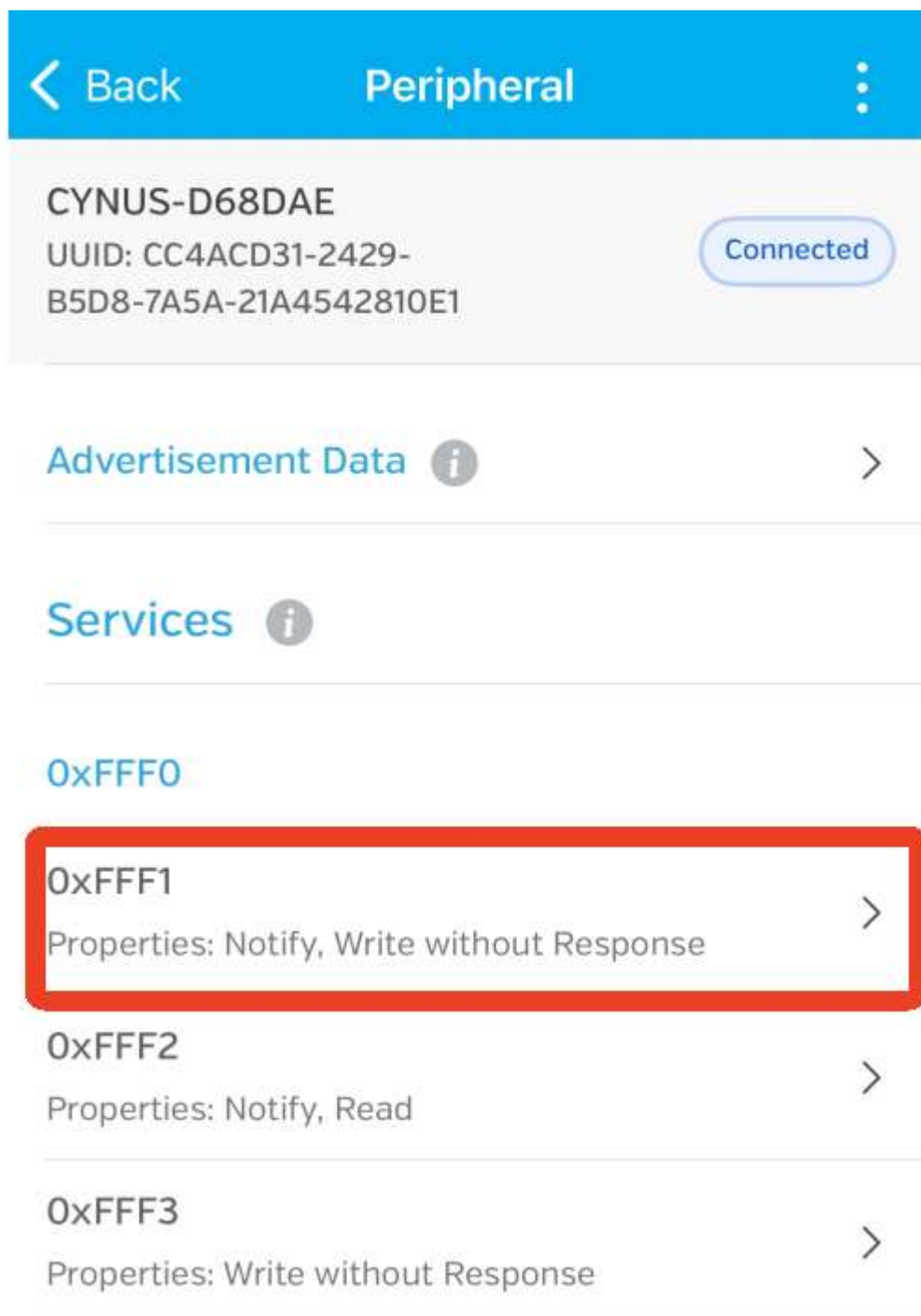


Cynus BLE protocol v3

1 notes

1:the protocol is base on Bluetooth LE 5.1, the Characteristic of Cynus' bluetooth which is used for communication is "FFF1"



2:the protocol is a UCI similar protocol, all the commands end with ‘**n**’.

2 commands

1: playing related commands

1:set internal engine on/off

when use external engine to control the robotic arm, you need to send "set internal engine off\n" first to disable the built-in stockfish, then each time you press the clock button on chessboard, Cynus will send you a position string along with a “get move\n” command.

otherwise the only the position string of the board will be sent to the you, without “get move\n”, in this case do not control the arm by “move AxB\n” command, because the built-in engine will control the robotic arm.

The built-in stockfish will be enabled automatically when the BLE connection gets broken.

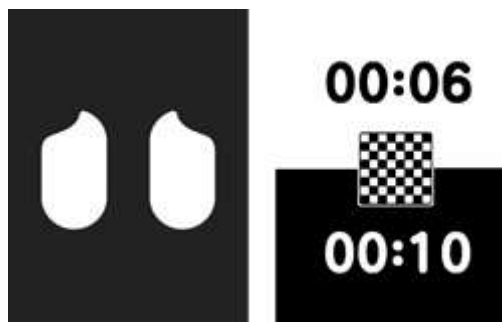
2: set flip board on/off

The robot plays the black chess pieces by default, “set flip board on\n” will flip the board, and the robot can play the white chess pieces.

It was set to off by default every time power up.

3: set time 10 / unlimited

when set time unlimited, emoji shows on the screen. otherwise, countdown timer shows on the screen.



4: get fen

By the default Cynus will push the current position to you when you connect to it by BLE. And push the position each time you press the clock button.

But you can still ask the Cynus to send you the current position by this command.

```
(ble_dev) PS F:\development\BLE> python .\ble_demo3.py
starting scan...
0 : 19:60:32:C4:86:58: CYNUS-326019
try to connect to device:19:60:32:C4:86:58
connected
# rev data: bytearray(b'promotions: ----\r\r fen: rnbqkbnr/pppppppp/8/8/8/8/PPPPPPPP/RNBQKBNR\r\n')
get fen
# You entered: get fen
rev data: bytearray(b'fen: rnbqkbnr/pppppppp/8/8/8/8/PPPPPPPP/RNBQKBNR\r\n')
```

The position string in red is sent out when the Cynus gets connected. And the position string is sent out on the command “get fen”

5: move AxBy

Send the move chess command to cynus, cynus will move the pieces.

other commands:

1: force reset

this command will reset the arm to the state which is ready for board scanning.



2: force fold

this command will fold the arm to the state which is ready for packaging.



3: force grab

Move the magnet in the gripper down to grab chess piece.

4: force release

Move the magnet in the gripper up to release chess piece.

5: get serial

get the serial number of cynus, cynus will return the serial string "serial 14d099f4901abd2d\n"

```
# You entered: get serial
rev data: bytearray(b'serial 14d099f4901abd2d\n')
```

6: get battery

get the battery's voltage of cynus. cynus will return the voltage of the cynus "battery 8.3v\n". there is a two cell li-po battery embedded, the voltage is 8.4V when full, and 7.4v when empty.

```
# You entered: get battery
rev data: bytearray(b'battery 8.2v\r\n')
```

6: get software version

Cynus will return its software version

```
# You entered: get software version
rev data: bytearray(b'App version: V1.0.0\r\n')
```

6: get hardware version

Cynus will return its hardware version

```
# You entered: get hardware version
rev data: bytearray(b'hardware version: V1\r\n')
```

3: example

ble_stockfish2.py is an example which use chess engine running on PC in Python to control the cynus.

Here are main steps of the program:

- 1: import bleak for BLE connection and import stockfish as chess engine.
- 2: search the BLE device whose name start with CYNUS and connect to it.
- 3: send to command "set internal engine off\n" to cynus to disable the built-in chess engine.
- 4: press the clock button on the chessboard to trigger a robot move. Robot will scan the board and send the position to your program.
- 5: then Cynus will send "get move\n" command to your program in sequence.
- 6: send the fen string to local chess engine and get the best move.
- 7: send the command "move xxxx\n" to Cynus. Cynus will move the chess pieces.

```
(ble_dev) PS F:\development\BLE> python .\ble_stockfish2.py
starting scan...
0 : 19:60:32:C4:86:58: CYNUS-326019
try to connect to device:19:60:32:C4:86:58
connected
# set internal engine off
# You entered: set internal engine off
rev data: bytearray(b'promotions: ----\r\n')
rev data: bytearray(b'fen: rnbqkbnr/pppppppp/8/8/8/8/PPPPPPPP/RNBQKBNR\r\n')
rev data: bytearray(b'get move\r\n')
cur_fen1: rnbqkbnr/pppppppp/8/8/8/8/PPPPPPPP/RNBQKBNR b - - 0 15
+---+---+---+---+---+---+---+---+
| r | n | b | q | k | b | n | r |
+---+---+---+---+---+---+---+---+
| p | p | p | p | p | p | p | p |
+---+---+---+---+---+---+---+---+
|   |   |   |   |   |   |   |   |
+---+---+---+---+---+---+---+---+
|   |   |   |   |   |   |   |   |
+---+---+---+---+---+---+---+---+
|   |   |   |   |   |   |   |   |
+---+---+---+---+---+---+---+---+
|   |   |   |   |   |   |   |   |
+---+---+---+---+---+---+---+---+
|   |   |   |   |   |   |   |   |
+---+---+---+---+---+---+---+---+
| P | P | P | P | P | P | P | P |
+---+---+---+---+---+---+---+---+
| R | N | B | Q | K | B | N | R |
+---+---+---+---+---+---+---+---+
best move d7d6
```